

HW 3 - Zeshui Song

4.6 6. Calculate the power dissipated in the $1\ \Omega$ resistor of Fig. 4.36.

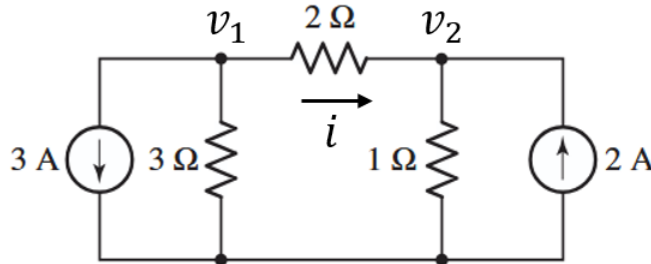


FIGURE 4.36 *Ref.*

Using nodal analysis at node 1:

$$[-3\text{ A}] = \frac{v_1}{[3\ \Omega]} + \frac{v_1 - v_2}{[2\ \Omega]} \quad \Rightarrow \quad -3 = \frac{5}{6}v_1 - \frac{1}{2}v_2$$

At node 2:

$$\frac{v_1 - v_2}{[2\ \Omega]} = \frac{v_2}{[1\ \Omega]} - [2\text{ A}] \quad \Rightarrow \quad -2 = \frac{1}{2}v_1 - \frac{3}{2}v_2$$

Solving the system of equations, we get $v_1 = -7/2\text{ V}$ and $v_2 = 1/6\text{ V}$.

The power dissipated by the $1\ \Omega$ resistor is then

$$P = VI = \frac{v_2^2}{R} = \frac{[1/6\text{ V}]^2}{[1\ \Omega]} = 0.02777\text{ W}$$

$P = 0.0278\text{ W}$

4.12 12. Using the bottom node as reference, determine the voltage across the $5\ \Omega$ resistor in the circuit of Fig. 4.42, and calculate the power dissipated by the $7\ \Omega$ resistor.

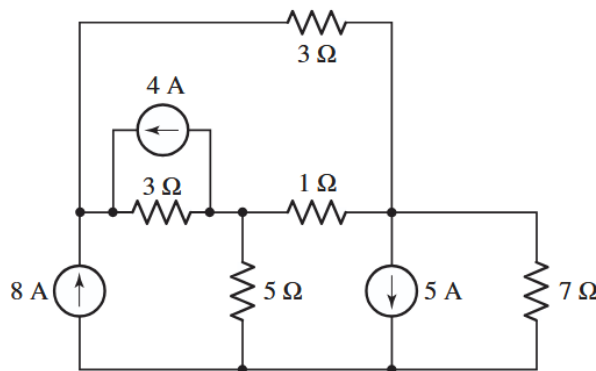
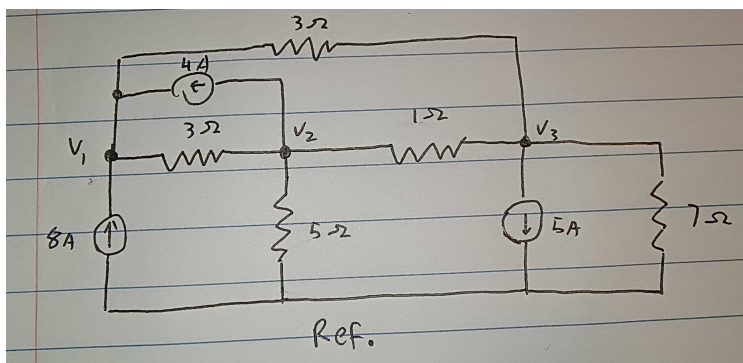


FIGURE 4.42

Redrawing the circuit slightly and labeling the nodes:



At node 1:

$$[8 \text{ A}] = \frac{v_1 - v_3}{[3 \Omega]} - [4 \text{ A}] + \frac{v_1 - v_2}{[3 \Omega]} \quad \Rightarrow \quad 12 = \frac{2}{3}v_1 - \frac{1}{3}v_2 - \frac{1}{3}v_3$$

At node 2:

$$\frac{v_1 - v_2}{[3 \Omega]} = [4 \text{ A}] + \frac{v_2 - v_3}{[1 \Omega]} + \frac{v_2}{[5 \Omega]} \quad \Rightarrow \quad 4 = \frac{1}{3}v_1 - \frac{23}{15}v_2 + v_3$$

At node 3:

$$\frac{v_2 - v_3}{[1 \Omega]} + \frac{v_1 - v_3}{[3 \Omega]} = [5 \text{ A}] + \frac{v_3}{[7 \Omega]} \quad \Rightarrow \quad 5 = \frac{1}{3}v_1 + v_2 - \frac{31}{21}v_3$$

Solving the system of equations, we get $v_1 = 26.733 \text{ V}$, $v_2 = 8.833 \text{ V}$, and $v_3 = 8.633 \text{ V}$.

The power dissipated by the 7Ω resistor is then

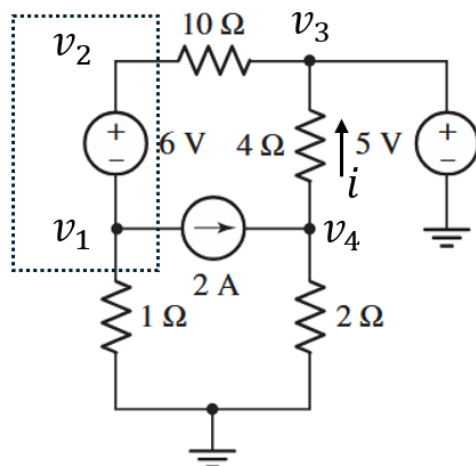
$$P = VI = \frac{v_3^2}{R} = \frac{[8.633 \text{ V}]^2}{[7 \Omega]} = 10.54 \text{ W}$$

$$\boxed{P = 10.64 \text{ W}}$$

and the voltage across the 5Ω resistor is

$$\boxed{v_3 = 8.633 \text{ V}}$$

- 4.20 20. For the circuit of Fig. 4.50, determine all four nodal voltages.



■ FIGURE 4.50

Using a super node around the voltage source at nodes 1 and 2:

$$v_2 - v_1 = [6 \text{ V}] \quad \Rightarrow \quad 6 = -1v_1 + v_2$$

$$\frac{0 - v_1}{[1 \Omega]} = \frac{v_2 - v_3}{[10 \Omega]} + [2 \text{ A}] \quad \Rightarrow \quad 2 = -v_1 - \frac{1}{10}v_2 + \frac{1}{10}v_3$$

At node 3:

$$v_3 = [5 \text{ V}]$$

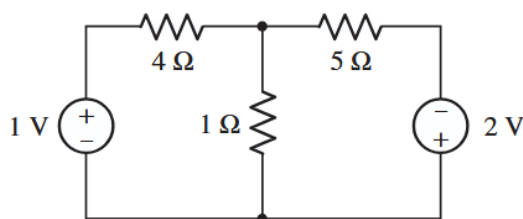
At node 4:

$$[2 \text{ A}] = \frac{v_4}{[2 \Omega]} + \frac{v_4 - v_3}{[4 \Omega]} \quad \Rightarrow \quad 2 = -\frac{1}{4}v_3 + \frac{3}{4}v_4$$

Solving the system of equations, we get

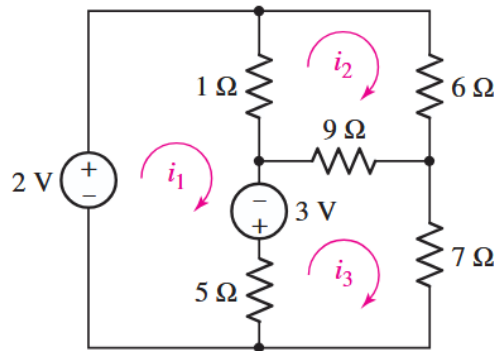
$$v_1 = -1.909 \text{ V}, v_2 = 4.0909 \text{ V}, v_3 = 5 \text{ V}, \text{ and } v_4 = 4.333 \text{ V}$$

- 4.32 32. Determine the currents flowing out of the positive terminal of each voltage source in the circuit of Fig. 4.60.



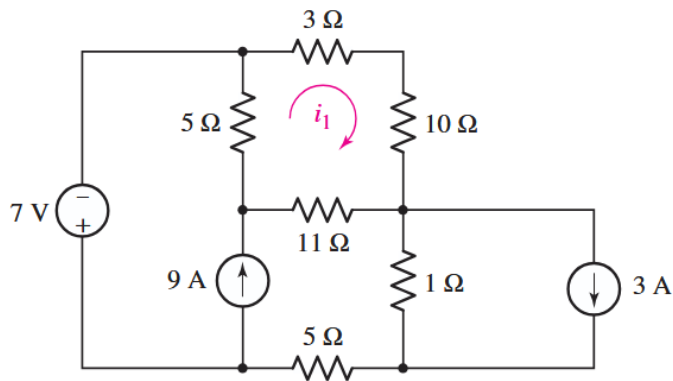
■ FIGURE 4.60

- 4.36** 36. Calculate the power dissipated by each resistor in the circuit of Fig. 4.63.



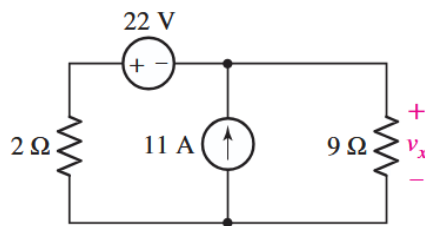
■ FIGURE 4.63

- 4.46** 46. For the circuit of Fig. 4.73, determine the mesh current i_1 and the power dissipated by the $1\ \Omega$ resistor.



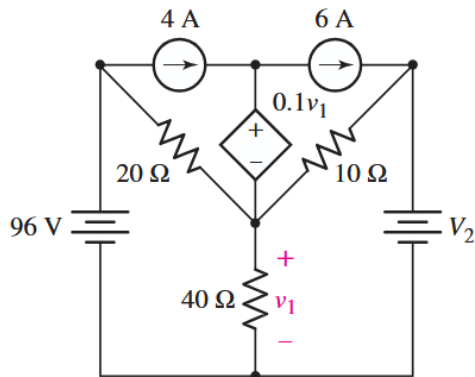
■ FIGURE 4.73

- 4.56** 56. Solve for the voltage v_x as labeled in the circuit of Fig. 4.82 using (a) mesh analysis. (b) Repeat using nodal analysis. (c) Which approach was easier, and why?



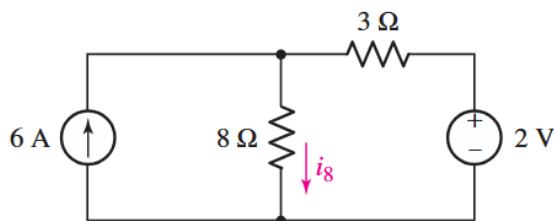
■ FIGURE 4.82

- 4.57 57. Consider the five-source circuit of Fig. 4.83. Determine the total number of simultaneous equations that must be solved in order to determine v_1 using (a) nodal analysis; (b) mesh analysis. (c) Which method is preferred, and does it depend on which side of the $40\ \Omega$ resistor is chosen as the reference node? Explain your answer.



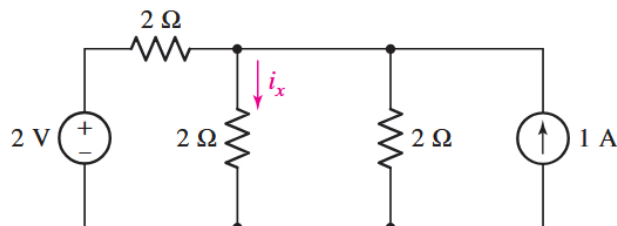
■ FIGURE 4.83

- 5.3 3. Considering the circuit of Fig. 5.48, employ superposition to determine the two components of i_8 arising from the action of the two independent sources, respectively.



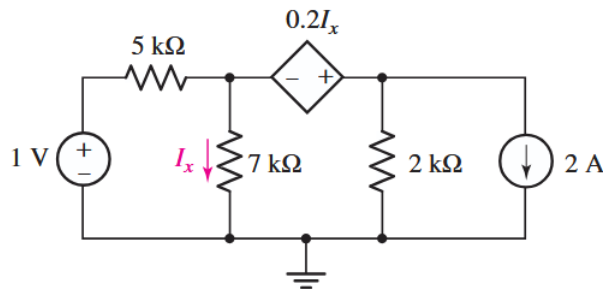
■ FIGURE 5.48

- 5.5 5. (a) Using superposition to consider each source one at a time, compute i_x . (b) Determine the percentage of i_x arising from each source. (c) Adjusting only the current source, alter the circuit to double i_x .



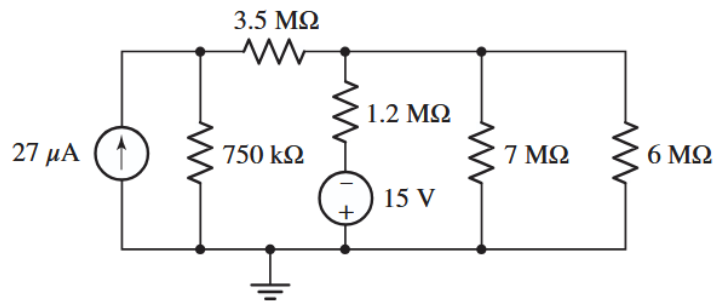
■ FIGURE 5.50

- 5.11** 11. Employ superposition principles to obtain a value for the current I_x as labeled in Fig. 5.56.



■ **FIGURE 5.56**

- 5.18** 18. (a) Using repeated source transformations, reduce the circuit of Fig. 5.62 to a voltage source in series with a resistor, both of which are in series with the $6\text{ M}\Omega$ resistor. (b) Calculate the power dissipated by the $6\text{ M}\Omega$ resistor using your simplified circuit.



■ **FIGURE 5.62**